DATA STRUCTURES • MINI PROJECT

RAILWAY RESERVATION SYSTEM

“Making every journey easier, one booking at a time.”

PROJECT SNAPSHOT	CORE IDEA
A menu-driven C application for train search, ticket booking, cancellation, and reservation display.	Use data structures to organize train and passenger records clearly and efficiently.

1. Abstract

Railway transportation connects people, places, and opportunities. A simple and reliable reservation process helps passengers find trains, check seat availability, book tickets, and cancel reservations without unnecessary manual work. This project proposes a **Railway Reservation System** developed using the C programming language and fundamental data structures.

The system maintains train details such as train number, train name, source, destination, and available seats. It also records passenger information and booking details. Through a menu-driven interface, the user can view trains, search for a route, reserve a ticket, cancel an existing booking, and display reservation information. Before confirming a booking, the program checks seat availability and updates the count after booking or cancellation.

2. Problem Statement

Managing train and passenger records manually can be slow and may cause mistakes in seat counts or reservation details. The project addresses this at a student-project level by creating a structured program that searches records, checks availability, and updates bookings consistently.

3. Objectives

- Organize train and passenger information in a clear format.
- Search trains using train number or source and destination.
- Book a ticket only when a seat is available.
- Cancel a valid booking and update seat availability.
- Display booking status and provide clear messages for invalid requests.
- Demonstrate practical use of structures, arrays or linked lists, and searching.

Design & Data Structures

4. Data Structures Used

Structure (struct): Combines related fields—such as train number, name, route, and available seats—into a single record.

Array: Stores a fixed collection of train or booking records, making it straightforward to access records by index.

Linked list (optional): Can store records dynamically when the number of entries may change during program execution.

Linear search: Checks records one by one to find a train or booking that matches the user's input.

5. Main Modules

MODULE	WHAT IT DOES
Train Details	Stores and displays train number, name, route, and seat count.
Train Search	Finds trains matching the requested source and destination.
Ticket Booking	Checks seats, stores passenger details, and creates a booking ID.
Cancellation	Finds an active booking, cancels it, and restores one available seat.
Reservation View	Displays booking ID, passenger details, train, route, and status.

6. System Workflow

01	**Open menu** Choose search, booking, cancellation, or view options.
02	**Find train** Enter train number or source and destination.
03	**Check seats** Confirm the requested train has available seats.
04	**Update record** Create or cancel the booking and adjust seat count.
05	**Show result** Display a booking ID, updated availability, or an error message.

Algorithms, Output & Conclusion

7. Basic Algorithms

Ticket booking: Read the train number and passenger details. Search for the train. If it exists and seats are available, create a booking ID, save the reservation, decrease the available-seat count by one, and show confirmation. Otherwise, show an appropriate message.

Ticket cancellation: Read the booking ID and search for an active reservation. If found, mark it cancelled and increase the corresponding train's available-seat count. If no active booking is found, report that the booking could not be located.

8. Expected Output

- Train list with route and available-seat details.
- Search results for the selected stations.
- Booking confirmation with a unique booking ID.
- Clear message if a train is invalid or no seats are available.
- Cancellation confirmation and updated seat count.
- Passenger reservation details and booking status.

9. Advantages

- Easy-to-understand menu-driven interface.
- Helps organize reservation records and seat counts.
- Makes basic search, booking, and cancellation operations easier to demonstrate.
- Provides practical experience with C structures and data structures.

10. Limitations & Future Enhancements

This is an academic prototype and does not connect to the real railway reservation network, live schedules, online payments, or real-time seat inventory. If file handling is not implemented, records may be lost when the program closes. Future versions could add file storage, passenger login, fare calculation, coach and seat selection, and a graphical or web interface.

11. Conclusion

The Railway Reservation System demonstrates how basic data structures can support a familiar real-world task. By combining train search, seat checking, booking, cancellation, and reservation display, the project reinforces structured programming and record management. It can be expanded gradually into a more complete reservation application.

PROJECT TAKEAWAY

A small, practical project that connects data structures with a real-life booking process.